

CHAPTER 2

LITERATURE SURVEY

2.1 LITERATURE REVIEW

This chapter explores the diverse methodologies employed by various authors in predicting DR and glaucoma, highlighting their key findings and approaches. It is divided into six sub-sections: Image Analysis of Retinal Fundus Characteristics based on DL methods, glaucoma Screening based on Fundus Images using DL methods, DR disease based on DL methods for classification, and ML and DL methods for glaucoma disease prediction.

2.1.1 Image Analysis of Retinal Fundus Characteristics based on Deep Learning Methods

González-Gonzalo et al. [2] discussed a deep visualizing procedure enhances augmentation of the visual evidence on DL classification problems in medical imaging. The approach proposed here relies on the score obtained with a classifier learned only with image-level annotations to unveil anomalies. A specific visual attribution technique is used to obtain the first visual evidence of the prediction for each image. This enables the localization of anomalies, which are subsequently eliminated by selective in painting. The author used this technique to score color fundus images of two retinal disorders: DR and Age-related Macular Degeneration (AMD). It assesses both qualitatively and quantitatively the visual evidence produced and the effectiveness of weakly-supervised localization of various AMD and DR issues.

Rodrigues et al. [3] detailed the vascular structures of the retina indicated above include crucial information regarding the identification with evaluation of vision problems including DR, glaucoma, and age-related macular degeneration. Retinal vascular segmentation is typically done interactively or manually, which

adds time and increases the possibility of human mistake. ELEMENT, a novel multi-modal framework for vessel segmentation, was implemented to address the problem. For pixel-based classification and feature extraction, this framework makes use of region growth and ML. The information is efficiently transmitted across the pixels during the categorization stage. The effect of such implementation enhances segmentation throughput and at the same minimizes possible disparities.

Li et al. [4] addressed the significance of automatic retinal vascular segmentation in Ocular Disease Diagnosis (ODD) and prevention. Every pixel is treated identically by the DL retinal vascular segmentation algorithms that are currently in use. Nonetheless, particularly in thin vessels, the multi-scale vessel structure plays a crucial role in influencing the segmentation outcomes. A unique Fully Attention-based Network (FANet) technique was put out to solve the problem. Its foundation was the use of attention processes to aggregate the multi-scale data and adaptively develop rich feature representation. In particular, the semantic segmentation networks and the picture pre-processing process make up the framework. Additionally, the network integrates the U-Net with two different kinds of attention modules. In this lightweight path of bidirectional attention module, the weights of the feature maps are reforming based on the semantic relationship, which captures global context and reduces inter-instance variations. This results in the creation of the attention map by the two-way attention block through both the horizontal and vertical pooling mechanisms. Thus, the network collects contextual data from around the world from a set of pixels that refers to the same object category or from semantically similar places.

Zou et al. [5] emphasized how difficult it is to segment tiny retinal in fundus images that are smaller than two pixels wide. They suggested a local regression system to improve certain areas of the narrow vessels, particularly the thin ones, and a new multi-label classification technique that makes use of this

plan to efficiently segment the vessels. Using the vessel shapes from the actual images, a local inverse regression model is first used to infer the multi-labels. Furthermore, it improves diabetes images for multi-label classification using a CNN model. To obtain a better promising small vessel and to capture the entire retinal vessel map, it is necessary to employ the transfer learning of the local regression to decode the preceding multiple-label data into binary labels. The outcomes of the trial have shown how well our technique works for segmenting retinal vessels.

Lin et al. [6] explained how a precise diagnosis of fundus disease can help patients receive tailored treatment strategies and effectively prevent the disease from getting worse. Fundus classifications of images may include one or more abnormalities in the fundus image. The authors suggested two novel multi-label classification networks, first one is MCG-Net (Multilabel Convolutional Graph Network), second is MCGS-Net (Multilabel Convolutional Graph Network with Self-supervised Learning). In this case, the multi-label fundus images pertinent information is captured by the graph convolutional network, and the network's capacity to generalize is improved by self-supervised learning, which learns more unannotated data. It can greatly increase the generalization capacity and classification performance of multi-label fundus image classification.

Wang et al. [7] explained two specific aspects of the graph-based approach used to separate retinal Artery/Vein (A/V) are the method outlined here for vascular tree unwrapping and vessel type classification. Current methods treat them as two different activities and solve them mainly with hand-engineered features (such as color/ thickness) and heuristics (such as changing the direction of the vessel). As a result, the approach of developing distinct characteristics and models for the two tasks frequently results in the development of an imperfect A/V separation solution. In order to give more accurate deep features for a subsequent accurate A/V separation, a multi-task Siamese network was put

forward to learn both tasks simultaneously. It is suggested that a Convolution Along Vessel (CAV) be used, which convolves the fundus image in the vessel section to provide geometric features and visual features by indicating the direction of blood flow within the vessel. After that, two jobs are taught to the Siamese network in order to break down the vascular system into separate vascular trees. i) classify proposed vessel segments into A/V clusters based only on visual aspect features, and ii) use related similarity metrics. By contrasting geometrical and visual characteristics, the two groups are connected. Ultimately, two tasks outcomes mutually correct one another to achieve the ultimate A/V separation.

Annamalai and Nedunchelian [8] proposed new method known as Optimal Weighted Based Deep Artificial Neural Network (OWDANN) technique which resolved the issue by additional training time and provided low detection accuracy. Many could avert preventable blindness if these anomalies were identified and treated early to stop future development. Khanna et al. [9] proposed a CNN-Long Short-Term Memory (CNN-LSTM) technique. This method applied as a network ensemble to predict DR. this proposed model performs better than current classification models.

Zhang et al. [10] explained severity of retinal lesions result in irreversible visual damage, and retinal-related disorders are the primary cause of vision loss. For prompt treatment, automatic techniques for detecting retinal disorders from medical photographs are therefore crucial. DL can efficiently diagnose and find important biomarkers. However, a vast amount of highly qualified specialists is mandatory for manual diagnosis and interpretation of medical images. Cycle Generative Adversarial Network and Convolutional Neural Network (CycleGAN-CNN) techniques were used to achieve the goal with little training data, it was utilized to identify lesion locations and identify retinal disorders. Cycle-consistent images generated by the CycleGAN are more reliable and lifelike. The generator and the classifier worked together to differentiate the input image domain, while

the discriminator and the generator work together in an adversarial fashion to reach a local optimal solution. Pixel-adaptive convolutions and learnable residual features are combined to provide a unique res-guided sampling block. The res-guided sample blocks are used in the place of conventional convolution to create a res-guided U-Net functioning as the generator.

Bilal et al. [11] explained a new approach for the automatic detection belonging to DR from fundus. The authors integrated pretrained learning through U-Net design to improve both the segmentation and classification of DR features. Domain adaptation influences pre-trained prototypes to enhance model performance, particularly when working with limited datasets.

Jeong et al. [12] proposed the Ocular Axial Length (OAL) technique, which is a crucial characteristic of eyes and that is used to assess their health before surgery. When creating artificial lenses to replace damaged natural lenses, OAL estimation is also essential. However, a large, expensive benchtop optical setup is needed to detect OAL accurately. Fundus images, which are now easily taken with small cameras, can capture the intricate anatomical details of eyes and to guarantee the architecture designed freely. Post-processing which is distinct from the training process to achieve this visual interpretation and highlighted a specific area of interest for predictions. The visual outcomes of this method suggested a connection among OAL and the biological makeup of the eyes, opening the door for additional research.

Radha and Karuna [13] studied the vascular features such as vessel width and contrast are thin, and low contrast has been established as the correct diagnostic feature of DR within the described retinal fundus photographs. To replace labeling operations, sophisticated automatic segmentation techniques predicated upon the Deep CNN (DCNN) have been created. When it comes to fundus vascular segmentation, the methods that have been provided have performed fairly well. From the DCNN model, the UNet design that has been

established for vessel segmentation then provided impressive results. An encoder/decoder system reduction, many coils, and poor feature fusion are addressed by Depthwise Parallel Attention method in the Existing UNet framework (DPA-UNet) was used. This lessens the information loss brought on by traditional down sampling and convolution approaches. An alternate attention method is suggested for the present UNet's upsampling path and spatial information processed by the encoder-decoder. DPA-UNet comprises convolutions that are performed by escalating the number of multiplications. These difficulties finally lead to the degradation of segmentation performance due to information loss about thin and low contrast vessels.

Das and Nayak [14] proposed a Fuzzy Joint Attention-Guided Network (FJA-Net) which was implemented to check for glaucoma phases in order to address the problem. The contributor additionally incorporates a Fuzzy Joint Attention Module (FJAM) on top of a backbone to acquire full feature correlations along the pertinent channels and spatial placements. This module consists of a local-global channel and a spatial attention block, each of which is followed by a fuzzy layer to reduce feature representation uncertainty. Critical areas of the fundus pictures are used to impart the FJAM fine-grained, and stage-specific characteristics.

Das et al. [15] explained chronic glaucoma, a degenerative eye condition left untreated, which can result in irreversible visual damage. Therefore, preventing vision loss and creating a more effective treatment strategy depend heavily on early glaucoma detection. However, significant interstage similarities, minor changes in lesion size, and the inclusion of irrelevant characteristics make it difficult to reliably define the glaucoma phases. Furthermore, fundus images include a great deal of ambiguous information that traditional CNNs are unable to adequately capture.

Islam et al. [16] pointed out DR has an inappropriately segmented sections which are improved and removed using morphological opening, adaptive histogram equalization. Patients with diabetes or other conditions showed much more optical nerve growth than healthy individuals. The author used a CNN model to train the classifier to achieve classification. Based on the classification-centric architecture of CNN, each class is evaluated with suppression, excitation, and interference layers. It also determines the diabetic images through a convolution and pooling layer design for classification between two classes.

Nazir et al. [17] discussed the minimization of the occurrence and worsening of vision loss necessitates the identification of DR and its treatment in its early stages. DL techniques now outcompete other methods with regard to DR diagnoses using retinal pictures. According to the study design, only DL-based studies stood comprised, and the examine technique was grounded on the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) strategies. Another way to reduce misdiagnosis and improve efficiency is to use DL-based automated algorithms to evaluate DR using color fundus pictures. Significant advantages are provided by a DL based automated method to lower screening expenses, improve early treatment outcomes, and increase access to healthcare.

Ali et al. [18] discussed glaucoma affects the OD and results in visualization harm in its advanced stages, while DR damages the retina and Diabetic Macular Edema (DME) is caused by fluid buildup in the macula. Nevertheless, because the disease progresses slowly, it exhibits few symptoms in its early stages, making disease detection challenging. For the detection and screening process to be supported in its early stages, a completely automated system is necessary. The study used Fuzzy K-means (FKM) clustering and the Fast Region-based CNN (FRCNN) technique towards offer an automated approach for disease segmentation and location detection. First, FKM clustering is

used to segment the annotated images, and FRCNN is employed in place of localization training. The segmented area and the ground truth are then compared using a cross-over function.

Kang et al. [19] highlighted that the purpose of this assignment was to show how ML techniques can be used to segment and categorize DR. Retinal Fundus pictures in two dimensions were employed. The authors introduced a new automatic region-growing methodology based on clustering. Improving classification accuracy by performing a combined hybrid-feature dataset used a data fusion technique. As of right now, automated detection and segmentation have played a significant role in providing crucial data for determining the phases of DR.

Son et al. [20] used DL techniques on retinal imaging to identify cardiovascular risks and ocular conditions. Moreover, renal function deficiencies were linked to structural and microvascular alterations in the retina. The author's goal was to use retinal fundus pictures to create and assess a DL model for early renal function impairment detection. To identify decreased renal function from the pictures, a DL model was built. This DL model makes it possible to use retinal fundus pictures to identify early renal function degradation.

Kim et al. [21] sought to create and assess DL models that filter retinal fundus pictures for a variety of aberrant features. While the conception of the lesions was subjectively examined, the DL techniques at the maximum harmonic mean assessed then contrasted with the efficiency of experts. The DL methods with guidance of region presented consistent act for recognizing many outcomes in macula-centered fundus.

Lee et al. [22] proposed AI based diagnostic in retinal images, such as fundus photo to differentiate between retinal disorders. Three CNN methodologies: ResNet50, VGG19, and Inception v3 were used to classify

retinal pictures. Additionally, a number of dense (completely connected) layers' performances was contrasted. These findings showed that the AI technology based on fundus images can be used to diagnose retinal disorders.

Wu and Cheddad [23] developed a framework for automatic fundus macular image augmentation in order to eliminate complex picture deterioration and enhance low-quality fundus images. A unique DL-based technique that automatically improves poor retinal fundus images with complex degradation was proposed by the author. It was utilized to create data augmentation techniques that replicated key elements of retinal picture deterioration and to suggest a CNN architecture that could be tailored to improve Low Quality (LQ) images based on the type of degradation. Complex degradation in fundus images, however, may result in expensive patient reexaminations or incorrect clinical diagnoses.

Thanki [24] described an image of the retinal fundus serve as a crucial foundation for evaluating various retinal conditions at early in diagnosis. The author suggested a novel DL-based method that automatically enhances subpar retinal fundus photos with complex deterioration. The performance of the original fundus pictures is then compared to these locations. The most advanced pre-trained models (AlexNet, GoogleNet, Resnet50, and VGG19) were used in both transfer DL and CNNs. Particularly when employing AlexNet/Resnet50, the section of the other regions shows greater predictive value than the original fundus image.

Aurangzeb et al. [25] discussed a common type of chronic disease, and it is a problem that will continue to become a global epidemic soon. The measurable examination likewise uncovers that those with diabetes are at a higher gamble of creating glaucoma than those without this infection. In the event that glaucoma doesn't show early enough when analysis is conceivable, the condition prompts visual impairment. The creator utilized a DNN to make and assess variety of retinal fundus pictures and classify glaucomatous retinal pictures as a component

of a better computer-helped emergency procedure. A DNN extracts the fundus retinal image's deep features, which are then classified and analyzed using a variety of ML classifiers. According to experimental data, the combination of a logistic regression-based classifier and a DNN performs better than any other glaucomatous triage system now in use, increasing classification specificity, sensitivity, and accuracy.

2.1.2 Glaucoma Screening Based on Fundus Images Using Softcomputing Methods

Shinde [26] explained that vision loss may result from glaucoma, AMD, and chronic DR. Therefore, in order to avoid irreparable vision loss, early detection and treatment are essential. Imageries of the fundus retina have been used comprehensively to aid in the detection of these conditions. AI methods have therefore been frequently used to get around these limitations. It was then discovered that CNN and SVM models were extensively utilized in DL and ML approaches, respectively. Addressed the shortcomings of the current approaches, though, and offered ideas for new approaches to precisely identify eye conditions.

Ajitha et al. [27] mentioned a neurodegenerative eye condition called glaucoma is brought on by elevated intraocular pressure in the retina. Secondly, it is a leading cause of blindness, and unfortunately, total blindness can occur if the patient does not receive an early diagnosis. This fundamental problem creates a strong demand for efficient systems that do not require sophisticated human resources, large-scale equipment, or excessive time. Using retinal fundus images, the author suggested an offline CAD method for glaucoma. Application was developed using image processing, ML, and DL techniques. Region of Interest (ROI) detection stands using bright point technique, while input image validation is done using the Le-Net structure test. The independent parts of the suggested CAD system design can be modified to address various glaucoma diagnosis and

categorization issues. The technique has been proved to be quite precise and dependable once the method was trained on numerous datasets.

Phene et al. [28] explained glaucoma, one of the foremost reasons of blindness and irreversible image loss in the globe, is brought on by elevated intraocular pressure inside eye. For the purpose to avoid vision loss, early and precise diagnosis is essential. Glaucoma diagnosis by hand is a difficult procedure that calls for years of experience and training. For the automatic diagnosis of glaucoma, the author used a reliable and accurate CNN-based technique. A 13-layer CNN is effectively trained using this dataset to mine significant features, which are subsequently evaluated and classified as either normal or glaucomatous.

Shanmugam et al. [29] studied results those were compared with eye care providers dataset to identify features of the Optic Nerve Head (ONH) based on recommendations made by Glaucoma Specialists (GSs) and develop DL methods to predict features that can be identified from color fundus images. A retrospective dataset of 86,618 photos that were evaluated for glaucomatous ONH (GON) was used to train a DL technique. Referable GON can be detected by a DL approach trained solely on fundus pictures, which has a greater sensitivity and similar specificity to eye care providers. On a separate dataset containing diagnoses derived from a comprehensive glaucoma workup, the system continued to perform well.

Ovreiu et al. [30] proposed a consistent physical issue to the optic nerve that causes fractional or complete vision misfortune is the sign of glaucoma. The primary driver of this condition is raised intraocular strain within the eye, which can hurt the optic nerve. Fundus photography offers significant bits of knowledge on eye wellbeing. This study recommends utilizing fundus pictures to gauge the CDR to analyze glaucoma. To see whether glaucoma is available, estimations are taken of the optic circle and OC. In this method, taking apart the OD is the underlying move toward diagnosing glaucoma. The CDR proportions of the

photographs that are most often used to assess glaucoma still up in the air. An irregular backwoods RF classifier was then used to sort the glaucoma photographs as per the CDR values. Several approaches, containing original U-Net, decomposable fully U-Net, and decomposable U-Net, are applied to assess the outcomes of the suggested strategy.

Hemelings et al. [31] discussed glaucoma symptoms those don't show up until the disease is in its most severe stages and the patient has already lost a large portion of his vision. Regular eye examinations are necessary because glaucoma detection is essential for preventing blindness. However, glaucoma may not require these tests to be diagnosed. CNNs have recently been employed in ophthalmology to identify certain eye conditions, like glaucoma, because to their demonstrated effectiveness in the early identification of several diseases. Using 201 layers of Densely Connected Neural Networks (DenseNet), which were first pre-trained on ImageNet using the ACRIMA retinal dataset, the authors suggest a revolutionary approach. Because this classification model was obtained, we might have cautious optimism regarding its potential for glaucoma early detection.

Cho et al. [32] have proposed dataset during the last ten years to use fundus images to detect glaucoma. They struggle to generalize to external test sets, but they perform remarkably well on internal ones. The majority of these are taught using information from a glaucoma clinic. Changes in fundus cameras, glaucoma prevalence data, and glaucoma ground truth ideas could all affect performance. The creators show that in a scope of troublesome situations, the recently distributed Relapse Organization for Glaucoma Reference (G-Chance) produces vital results. These outcomes affirm that a relapse model of glaucoma risk worked with reliable information from a solitary tertiary reference clinic has a serious level of generalizability.

Lee et al. [33] created and assessed a DL ensemble technique to automatically classify glaucoma stages based on severity. The glaucoma patients

were categorized using the standard automated perimetry mean deviation value. The author created an ensemble system by merging multiple modeling results after modeling 56 CNNs with various properties. Compared to the best single CNN framework, which is the least nearby misprediction, the proposed method has a reduced false negative. The method of averaging several CNN models is superior to a single CNN methodology for glaucoma stage classification using fundus pictures. The ensemble technique is a practical clinical decision support system for primary care glaucoma screening because it provides reliable and consistent results with a relatively modest amount of data.

Alghamdi and Abdel [34] examined to see whether long-term variations in the Retinal Nerve Fiber Layer (RNFL) thickness predictions made by a DL methodology using fundus photos can forecast the onset of glaucomatous visual field abnormalities in the future. During follow-up, more fundus and Standard Automated Perimetry (SAP) photos were obtained. Repeatable glaucomatous abnormalities on SAP were considered to be conversion to glaucoma. RNFL consistencies can be anticipated from fundus pictures by a DL strategy (machine to machine, or M2M) prepared on OCT was utilized. Consolidated longitudinal endurance models were utilized to survey the prescient force of gauge and longitudinal variations in M2M's RNFL thickness gauges for the beginning of visual field anomalies. Longitudinal varieties in a DL technique's evaluations of RNFL consistency estimates in light of fundus pictures can foresee the opportunity of glaucoma transformation in eyes envisioned of having the condition.

Hemelings et al. [35] explained about the challenging and time-consuming to label spots in fundus image. CNN is another tool that can be utilized to detect glaucoma. But for the training phase to be useful, there must be a lot of labeled samples accessible. A programmed glaucoma symptomatic framework in view of three CNN methods with particular learning modes has been created to address

this trouble. Utilizing non-clinical information that has previously been labeled utilizing a prepared CNN strategy, the recommended model's presentation is stood out from that of an Ophthalmologist. Secondly, in view of two particular solo methods, a semi-managed system is fabricated and prepared using both named and unlabeled information. Investigational results utilizing RIM-ONE and RIGA datasets, and shows DL technique is effective for glaucoma prediction. They suggested a possible step towards automatic screening to discover patients in the primary phases of the disease.

Mahum et al. [36] deployed a way to enhance explainable DCNN in the domains of glaucoma revealing and Vertical Cup-Disc Ratio (VCDR), a significant threat issue. Additionally, opposite cropping mask (periphery crop policy) was used. It offers the first indisputable proof that glaucoma can be identified by DL from ONH outside of the fundus.

Hirota et al. [37] explained one of the eye conditions that is triggered by elevated intraocular fluid pressure. Preventing vision loss requires meticulous screening and retinal image detection since glaucoma develops gradually and shows up in later stages to use DL-based feature extraction to identify glaucoma early. Retinal fundus pictures are utilized to prepare and test the conveyed approach. Moreover, the Local Binary Patterns (LBP) and Speeded-Up Robust Features (SURF) descriptors remain utilized to separate surface highlights, and Hoard is utilized to extricate low-level elements. Moreover, CNN is utilized to ascertain significant level qualities. To find the most delegate highlights, they likewise utilized an element choice and positioning technique known as the Mann-Whitney Ranking (MR) procedure.

Li et al. [38] addressed about using fundus photos with a small sample size to assess the CNN model's accuracy in glaucoma recognition using split color channels with three primary colors RGB. Using fundus photos with a modest sample size, CNN demonstrated the value of the green digital filter in building

CNN models for automatic glaucoma discrimination. According to the current findings, preprocessing is a vital period in the CNN model-building procedure for the diagnosis including numerous retinal disorders using color fundus photos.

Singh et al. [39] prescribed the impact of combining retinal images with patient health data is evaluated using a proposed Deep Residual Neural Network (DRNN) architecture to automatically detect GON from color fundus images. It assessed the model forecasts qualitatively along with conducted occlusion testing. An impact of including medical record information in the final classification was then evaluated. Furthermore, occlusion testing revealed that the most crucial elements for the discrimination of GON were the neuroretinal rim region and the sites of superior or inferior Retinal RNFL abnormalities. It could help Physicians and the general population with widespread glaucoma screening and swiftly, effectively, and affordably suggests the specialist a subsequent view on the identification of glaucoma.

Singh et al. [40] used embedded CAD systems that employed ML for classification, then feature reduction and selection by nature-inspired computing. The Bat Algorithm and Binary Cuckoo Search are highly successful at predicting glaucoma, according to the results of performance tests conducted on two representative publically accessible datasets. Two popular and openly available datasets were employed to assess the categorization accuracy of the suggested technique. In order to help Ophthalmologists diagnose glaucoma disease more rapidly and economically, this work offers an automated primary glaucoma screening method that makes use of quantitative analysis of fundus images.

Thanki [41] represented the use of heavy machinery and highly skilled workers in traditional glaucoma diagnosis methods makes it impractical to evaluate large groups of people. It is now possible to automatically extract features using DL approaches, something that was not achievable with conventional methods. Given the numerous related issues, such as the dearth of labeled data, the

expense and complexity of creating glaucoma fundus photographic datasets, and the need for specialized hardware, it evaluated the effectiveness of DL methodologies that are trained to identify glaucoma from fundus images and techniques. In the long run, this automated diagnosis approach could save Ophthalmologists' valuable time and human labor.

Medeiros et al. [42] discussed diabetes is a severe public health concern that is on the verge of becoming an epidemic on a global scale. It is a common chronic illness. Compared to people without diabetes, those who have the condition are more likely to acquire glaucoma. Blindness could ensue from glaucoma if it is not detected in time. In order to generate and assess color retinal fundus images and categorize glaucomatous images, the authors suggested an intelligent computer-aided triage method that makes use of a DNN and ML. A DNN used to obtain the fundus retinal photos deep characteristics, which are further categorized and examined using many ML classifiers.

Elangovan and Nath [43] revealed an uncommon picture characterization execution, CNN specifically has turned into a suitable choice for glaucoma identification from fundus pictures. Moving the streamlined loads from a current model accelerate and works on DNN preparing. To convey the best grouping results for glaucomatous and ordinary pictures, a profound outfit technique is planned utilizing the stacking gathering learning strategy. A two-step ensemble selection methodology is offered for determining the optimal configurations. A method of probability averaging was used to pool selected configurations. For the final classification, an SVM classifier was employed. Fundus images, however, take a lot of time and call for the knowledge of a qualified professional.

Yugha et al. [44] determined which areas of the retina are affected by glaucoma, doctors use precise testing methods. However, complex glaucoma testing procedures and a absence of sources sometimes result in delays in recognition, which could raise the prevalence of visual impairment globally.

Therefore, in order to overcome the encounters linked with physical practices, it is imperative to develop a practical, intelligent methodology that can correctly find the OD and OC lesions at an early stage. The Bi-Directional Feature Pyramid system modules of EfficientDet-DO continuously perform the top-down and bottom-up key point merger processes consuming the computed features from EfficientNet-B0. The confined parts of a glaucoma lesion and the corresponding categorization are predicted by the EfficientDet-DO.

Fan et al. [45] proposed the Data-Efficient Image Transformer (DeiT) and ResNet-50 approach are used by the Vision Transformer, which is based on the DL technique, to classify the images to perceive Primary Open-Angle Glaucoma (POAG) and towards guide each model's decision-making. The Ocular Hypertension Treatment Study (OHTS) compared the pinpointing interpretation and accuracy using fundus photos.

Shoukat et al. [46] addressed how people may be spared irreversible visual loss with early identification. Ophthalmologists use a variety of time-consuming, expensive, and skill-intensive techniques when manually assessing glaucoma. An automatic technique based on the DL approach was used to address the problem; it has a very high accuracy rate in detecting glaucoma in its early stages to find the patterns which was missed by the Physicians. The deployed methodology uses the data augmentation technique and the grey channels of fundus photographs to generate a huge dataset of flexible fundus to learn the CNN methodology. For prompt therapies, the suggested methodology might assist Physicians in making extremely accurate early-stage glaucoma diagnoses.

Kashyap et al. [47] suggested to detect and forecasting glaucoma before symptoms manifested. This DL system has been presented to analyse glaucoma images using the glaucoma dataset. The U-Net architecture incorporates pretrained transfer learning models to get the necessary outcomes when applying DL principles to the task of segmenting the OC. The DenseNet-201 Deep CNN

(DCNN) is utilized for feature extraction. To ascertain whether an individual has glaucoma, the DCNN method is employed. The main goal of this exploration line is to determine glaucoma in retinal images so that the condition can be identified in patients. It demonstrates that the new paradigm that has been suggested is more effective than the one that is being used at the moment.

Nayak et al. [48] described about diagnosis techniques rely on manually created feature engineering. Conversely, non-handcrafted feature extraction techniques are typically developed using gradient-based algorithms, which have serious drawbacks such as overfitting and require a higher sample size for efficient training. The issue was resolved using an Evolutionary Convolutional Network (ECNet) technique. The layers of convolutional, reduction, and Rectified Linear Units ReLU in the suggested approach make it easier to extract discriminative features. This benchmark uses maximizes the distance between different classes while reducing variation within each class, and this is used to train the ECNet.

2.1.3 Diabetic Retinal Disease Based on Deep Learning Methods for Classification

Albelaihi and Ibrahim [49] explained about the performance of the VGG16 and ResNet152V2 architectures which was used to provide the suggested EfficientNetB0 for analyzing diabetes prediction. Additionally, the public finance information is used to analyze and assess DL architecture classifications thoroughly. When given the option to choose from the models that are presented in DR, Additionally, there are few DL in the literature. The EfficientNetB0 performs superior than the other four suggested models, based the test findings. Diabetic eye disease is the prominent source of blindness globally, and diabetes is a severe community wellbeing problem.

Nazir et al. [50] described a novel approach that uses two primary steps: first, feature extraction and dataset preparation next, a proprietary DL-based

centroid model trained for the classification of retinal diseases is created. Furthermore, an annotation was created for the suspicious pattern where the accurate region is located in the diabetic retina images. After that, the other part of the proposed solution is demonstrated through the glaucoma images by training the core network model. Furthermore, the feature extraction method and the DENSENET-100 were used to improve assessment of localization and classification of retinal structure.

Bilal et al. [51] proposed and analyzed a two-step automatic DR classification method. By employing preprocessing and data amplification practices to improve image quality with size, it ascertains the low positive case rate in asymmetric disc and vessel detection systems. Likewise, it enables the separation of blood vessels and optic discs using two separate U-Net models. Additionally, a symmetric hybrid CNN model was constructed utilizing the Inception-V3 model grounded on TL to extract the OD and blood vessels. Similarly, pre-processing is performed to abstract and pick the most distinctive structures.

Islam et al. [52] proposed a DL framework to detect diabetic patients and make decisions based on retinal images. A CNN-based multi-level model, DiaNet was analyzed comparatively using a small dataset. Furthermore, it has been confirmed by medical experts to achieve an accuracy of maximum in the field of retinal work. Among these, the analysis focused on the discriminative ability of retinal images of diabetic patients in Qatari, which has been shown to be excellent. When comparing the performance of DiaNet with existing ML models, it has features based on clinical data to distinguish Qatari diabetic patients and controls in retinal images.

Qureshi et al. [53] proposed novel automatic method Active DL (ADL) framework based on a multi-layer structure to identify DR. Instead of manually extracting features, ADL uses a CNN model to extract features by generating a

feature set. The ADL-CNN framework proposes to select informative connections and images by simply learning complex retinal features using multiple ground truth labels from training samples. It allows clinicians to identify regions of interest in the severity of DR by providing a valuable predictive mask in critical eye samples and radiographic images.

Sarki et al. [54] recommended two scenarios for mild multiclass Diabetic Eye Disease (DED) and multiclass DE those were taken into consideration when describing the automatic classification method. By applying the suggested model to various datasets, Ophthalmologists use the labeled test to determine the findings. Likewise, two CNN prototypes already been skilled were used to conduct the experiments on ImageNet. Thus, Ophthalmologists save time and effort by creating an automated diagnostic system. Diagnosing DR is very challenging and time-consuming.

Goel et al. [55] explained a DL model used for hundreds of different kinds of DR fundus images to learn features and is trained using image metadata. It enables professionals to create models that can accurately or appropriately identify observed images. To accurately classify fundus images into severity categories, this research presented a DL model integrated with multiple classifiers. This model has demonstrated high precision subsequently being proficient on a dataset of snaps of DR from India. Negligence exacerbates eye disorders, causing more severe damage to the retina or retinal blood vessels, which eventually contributes to total blindness.

Sarki et al. [56] developed an automated Diabetic Eye Diseases (DED) categorization system, and a number of DED subtypes has been found. The detecting of multiple DEDs from retinal fundus is one of the most useful techniques. The classifier was verified using a number of eye fundus snaps collected from public databases and assessed by Ophthalmologists. It proceeds a

proportion of time and struggle to manually detect test images taken from a dataset using a CNN model.

Abbood et al. [57] recommended a process published through color fundus to improve the superiority of DR images by dropping noise and increasing contrast. Furthermore, shape cropping and Gaussian blurring were used to remove content from the dataset in two main stages: cropping the image, reducing noise, and enhancing contrast. The improved image shows better results in feature extraction and classification than without using the enhancement method.

Erciyas and Barışçı [58] suggested a DL-based approach to categorize glaucoma images. Additionally, information on DR were gathered out of several datasets to build the data table for the suggested approach. Furthermore, lesion detection and region of interest labeling are made possible by the suggested RCNN model. Additionally, utilizing TL and attention approaches, the acquired images are categorized as DR glaucoma images. However, diabetes has a significant impact on the economies of low- and middle-income governments.

De la Torre et al. [59] explained the DL algorithm performed exceptionally well when used to classify retinal images at different intensities for DR. Additionally, the pixel's contribution to the final classification is determined using the scores. The scores are then determined by splitting the observed output score in the neuron into two parts and putting forward a pixel-by-pixel scoring model. However, a preset optimization procedure for loss function minimization is used to change the model parameters.

Li et al. [60] described about deep ensemble models support by demonstrating superior performance with excellent toughness and generalizability in identifying DR and DME. The system assesses the generalization capabilities of the given algorithm on the publicly accessible Messidor-2 dataset and compares its performance to that of human specialists on a crucial dataset. Additionally, the

system analyzes how the size and number of input photos were utilized during training impact the model's performance.

Mohanty et al. [61] proposed an hybrid network model combining two DL architectures and using VGG16 and XGBoost models for DR detection classification is proposed. An unbalanced image class distribution was revealed and solved by appropriate balancing techniques based on the datasets. Additionally, the accuracy performance evaluation of the models that are presented is shown. Similar to that, the suggested approaches are contrasted with the current ways to improve the dataset's features, demonstrating the DenseNet 121 network's most significant performance. The analysis's findings show that DR based on DL architectures can be diagnosed and classified early.

2.2 ML and DL Methods for Glaucoma Disease Prediction

Thakur et al. [62] depicted DL models for OHTS. Onset of an anomaly with the subject is selected for review in order to validate the anomaly and enable different analyses by the endpoint group. 66,721 fundus photos were incorporated to learn and authorize the DL model, while remaining fundus images were used for validation. In addition, DL framework can very truthfully forecast the glaucoma onset before disease begins.

Akter et al. [63] suggested few approaches to OCT images and analyze various features to aid in the present diagnostic assessment of glaucoma. Utilized the four most crucial aspects to train the ML algorithm after statistically analyzing the features. The classification accuracy of two ML techniques, DL and Logistic Regression (LR) for automated glaucoma detection was compared.

Madadi et al. [64] revealed that progressive weighting technique reduces negative information transfer while accurately transferring knowledge from the input area to the destination field. Low-quality encryption is used to rectify the source and target distributions. Three scenarios serve as the foundation for Trained

Glaucoma Area Adaptation (GDA): glaucoma exclusion time with abnormalities in OD and visual field, glaucoma OD with abnormalities, and glaucoma without abnormalities in the OD or visual field.

Aljohani and Aburasain [65] introduced a system employing CNN (ResNet50, VGG-16) as a hybrid glaucoma detection architecture. They presented an advanced computerized system that uses clinical imaging data to accurately identify glaucoma by combining machine learning, CNN, and image processing. The model independently analyzes pre-processed retinal pictures and integrates the predictions with post-processing criteria to provide a thorough evaluation of the effects of glaucoma. Future developments in vision therapy may benefit from the use of DNN and ML in clinical image processing.

Raju et al. [66] had given a framework for glaucoma prediction using Electronic Health Records (EHRs) of various hospitals and health centers in the United States by using several ML and LR approaches, as well as chronological data processing. Predictive analysis of the complete dataset is carried out using four distinct ML classification techniques. The result indicates that pre-glaucoma patients can be identified using ML techniques, even before clinical symptoms appear, based on EHR data for early intervention and preventive care.

Pham et al. [67] used visual field and OCT pictures, and used DL to forecast the future visual field. The ResNet-50 learner was operated to examine image facts. Future scenes are predicted using a Long Short-Term Memory (LSTM) network model combines image attributes through the prior visual effects. The proposed method implemented weighting techniques to detect noisy data when weight loss is applied. However, collecting continuous visual field data is difficult, time-consuming, and may yield inconsistent results depending on the patient's condition.

Thakur and Juneja [68] proposed the methods like Xception, Inception, DenseNet, and ResNet which are used to perform comparative analysis based on cutting-edge DL approaches. Additionally, based on the suggested approaches, the comparison was done by utilizing the metrics like accuracy, recall, and precision. In order to provide portable glaucoma diagnostic tools, it was utilized to assess diabetic retinal pictures and forecast glaucoma. The diagnosis made by CAD systems is performed better in the presence of lighting interference, however, the imaging method was used to reduce the diagnosis time.

Huang et al. [69] proposed a probabilistic DL multimodel to predict glaucoma. Eighty percent of the fundus images and visual fields were used as the training dataset for the DL model, with the remaining twenty percent put aside for testing. Furthermore, a separate dataset was used to validate the model. A deterministic CNN model and the probabilistic DL model's performance were contrasted. CNN models that were probabilistic and deterministic performed similarly. However, the probabilistic models showed less confidence in their decision.

Sułot et al. [70] evaluated the utility of ONH in face images obtained by Scanning Laser Ophthalmoscopy (SLO) during conventional posterior segment OCT imaging. The potential utility is demonstrated by identifying regions of limited data that differentiate glaucoma patients from healthy controls using a clustering DL approach. Similarly, performance recognition and consistent accuracy were achieved based on cross-validation of SLO images in an ensemble of DL methods. Furthermore, this study showed that glaucoma images obtained with a diabetic retina dataset outperform all tested classifiers.

Badah et al. [71] represented on the Resnet-152 model and uses an SVM model based on several ML classifiers was used to diagnose human eye infections caused by glaucoma. The suggested approach was applied to the ophthalmological

cognitive recognition dataset to detect and enhance DR. Furthermore, the proposed image identification tasks have extensively used the DL model in recent years.

Juneja et al. [72] recommended a deep linkage model which was used to predict glaucoma in the Classification of Glaucoma Network (CoG-NET). Deep networks' capacity aimed at self-learning allows for quick and automatic diagnosis procedures. Furthermore, the presence of the OD and OC were identified by focusing on the corresponding functional activation graphs of the proposed network. Moreover, it was determined only in retinal images for glaucoma screening and glaucoma diagnosis.

Singh and Khanna [73] acknowledged as the first to provide a revolutionary multi-model method to glaucoma prediction, initially creating five multi-model illustrations of each widely available ranking databank. From the dataset, thirty-six key features were found. By combining these extracted features, 180 features were created for every dataset. A qualified Ophthalmologist receives the combined results of these classes. To prevent irreversible visual loss, glaucoma must be diagnosed and treated early.

Jesi et al. [74] presented the Deep Optimised ML Strategy (DOMLS), a novel method for early glaucoma diagnosis. This method captures gender and produces more accurate results by using a novel optimization logic known as Modified K-Means Optimization Logic (MkMOL). The article's main goals were to diagnose glaucoma in its initial junctures and use OCT pictures to give individuals valuable answers. Because of its structural consequences on the eyeball, this type of glaucoma is typically diagnosed with a complex cause.

Hussain et al. [75] created a multimodal DL methodology for predicting the progression of glaucoma that combines an LSTM network with a CNN. The suggested approach combined historical multimodal inputs with future photos synthesized by a GAN to anticipate VF changes 12 months after the initial visit.

This work offers important new information about the possibility of early glaucoma progression detection using synthetic follow-up OCT pictures.

Oh et al. [76] attempted to create an explanation system for a particular forecasting and an ML prediction model for glaucoma diagnosis. Clinical data from the patient's visual field test, fundus photography, general evaluation with IOP measurement, and RNFL-OCT test were provided for the feature selection process.

Oguz et al. [77] described DL-based architectures that use an automatic computerized method based on the segmentation of manually generated features in fundus images. Experts are interested in using a model that integrates raw fundus images with DL and ML techniques to diagnose glaucoma. Impaired vision and damage of central hallucination may result from the ailment's severity and irreversible damage to the optic nerve.

Afreen and Aluvalu [78] demonstrated how explainable AI is being used for glaucoma prediction analysis. AI that enables humans to understand AI predictions and decisions is known as Explainable AI (XAI). In contrast, the ML "black box" theory holds that even the AI's creator is unable to explain the reasoning behind some of its judgments. Manual glaucoma screening techniques take a lot of time and effort. However, DL-based glaucoma detection techniques increase speed and accuracy while reducing the requirement for manual labor.

Juneja et al. [79] used fusion framework of ML and DL procedures intended for glaucoma classification, consuming two popular imaging modalities, fundus imaging, and OCT, to assess whether an eye is normal or glaucomatous. It also evaluated the anticipated intraclass variability and identified the optimal predicted range to compare with existing techniques. The suggested approach may be helpful for OCT image-based glaucoma screening. Manual glaucoma diagnosis has several obstacles, including human error and time loss.

Wang et al. [80] developed a DL methodology to forecast, from a baseline visit, the likelihood of glaucoma surgery over a range of time horizons. Clinical and demographic characteristics, as well as spatially orientated visual field and OCT data, were all inputs into the model. Within defined time frameworks, the DL technique can efficiently identify eyes requiring treatment for recurrent glaucoma. As the time horizon for surgery predictions rises, predictive performance declines. Patients are identified through a referral process to a glaucoma expert for surgical examination when predictive models are used in a clinical context.

Mehta et al. [81] combined clinical and demographic data with multiple DNN trained on color fundus images along with macular OCT volumes as multimodel. The significance leads different features cutting-edge the diagnosis of glaucoma was assessed using interpretive ML techniques. Additionally, it evaluates a sample of individuals who were later identified with glaucoma but were not diagnosed on the day of imaging.

Jibhakate et al. [82] evaluated two distinct transfer learning methods and discussed the early identification of glaucoma. ML methodologies were used to quantify the morphological markers of glaucoma, which indicate when anomalies will begin to appear. The screening process for glaucoma is expensive, time haunting, and prone to human error.

Kumar et al. [83] centered on the CNN feature extraction and segmentation method for glaucoma disease detection. After that, fundus are cleansed in terms of CLAHE and histogram equalization procedure. Glaucoma primarily affects the nerves in the eyes, and if it is bad enough, it can cause irreversible blindness.

Juneja et al. [84] demonstrated a glaucoma proficient AI system that develops OD and OC segmentation. In a DL architecture designed to automate the identification of glaucoma, CNN is the central component. Glaucoma, an eye

disorder brought on by high IOP that harms the optic nerve in the retina, This is most frequent cause of lifelong blindness.

Sreng et al. [85] used an SVM technique, in which feature descriptions are generated by an ensemble of pre-trained deep CNNs. The finest decision for OD segmentation, according to examination of the techniques on five datasets with 2787 images, is a blending of DeepLabv3+ and MobileNet. Though, screening of retina is time-consuming, more skill labors, cost, and there are currently inadequate eye consultants available.

Prananda et al. [86] recommended the two stages one is pre-treatment phase and the second on is glaucoma classification phase. Additionally, the pretreatment procedure began with the removal of extraneous components like blood receptacles and the OD. Since they could cause problems throughout the analysis process, both components are considered for elimination. Lastly, the model would assistance in bettering the revealing and assessment of eye diseases.

Alwakid et al. [87] offered a DL model that contrast to conventional techniques, can reliably identify all five stages of DR. The suggested approach covers two scenarios. One scenario decides no picture augmentation, while the other utilizes an Enhanced Super-Resolution GAN (ESRGAN) in conjunction with the CLAHE filter technique. If there is no cure for DR, the goal of treatment is to preserve the patient's existing level of vision. Furthermore, DR is an indication of anomalies in the retinal vascular system, vision loss because of hemorrhage and scarring.

2.2 SUMMARY

This literature review concludes by examining the diverse methodologies and tools employed in developing diagnostic and treatment approaches for diabetic retinal and glaucoma. The review critically analyzes the advantages and disadvantages of different techniques, facilitating a clear identification of suitable methods for developing a secure and effective healthcare system that integrates innovative technologies. By synthesizing existing research, this review aims to provide a comprehensive overview of current methodologies and propose potential avenues for future research in this evolving field of ophthalmology.